perotac
MADE IN TAIWAN

MHI-Q

TWIN-PUMP BOOSTER SET WITH FREQUENCY INVERTER

2 × MHI pumps | 3~ 380–415 V | 50 Hz



2 × MHI · alternating / parallel operation

FEATURES

- Two MHI horizontal multi-stage stainless steel pumps on a common base frame
- Alternating operation: one pump runs while the other stands by, the pumps take turns
- Parallel operation: both pumps run together at peak demand
- Frequency inverter control in a stainless steel control cabinet
- Stainless steel suction and discharge manifolds with flanged connections
- Isolation valves on the suction and discharge of each pump
- Pressure tank, pressure gauge and pressure sensor on the discharge manifold

APPLICATION SCOPE

- Water supply and pressure boosting
- Apartment buildings, hotels and offices
- Air-conditioning and industrial circulation systems
- Washing and sprinkler systems

OPERATING DATA

Approved fluids	Clean water
Fluid temperature	0–90 °C
Max. suction lift	8 m
Max. pressure	10 bar
Protection class	IP54
Power supply	3~ 380–415 V 50 Hz
Number of pumps	2 (alternating or parallel)

MATERIALS

COMPONENT	MATERIAL
Casing	AISI 304 stainless steel
Impeller	AISI 304 stainless steel
Shaft	AISI 304 stainless steel
Mechanical seal	Silicon carbide / graphite, resistant to 120 °C
Manifolds	Stainless steel
Control cabinet	Stainless steel

OPERATING MODES

Alternating One pump runs, the other stands by; the two pumps take turns.

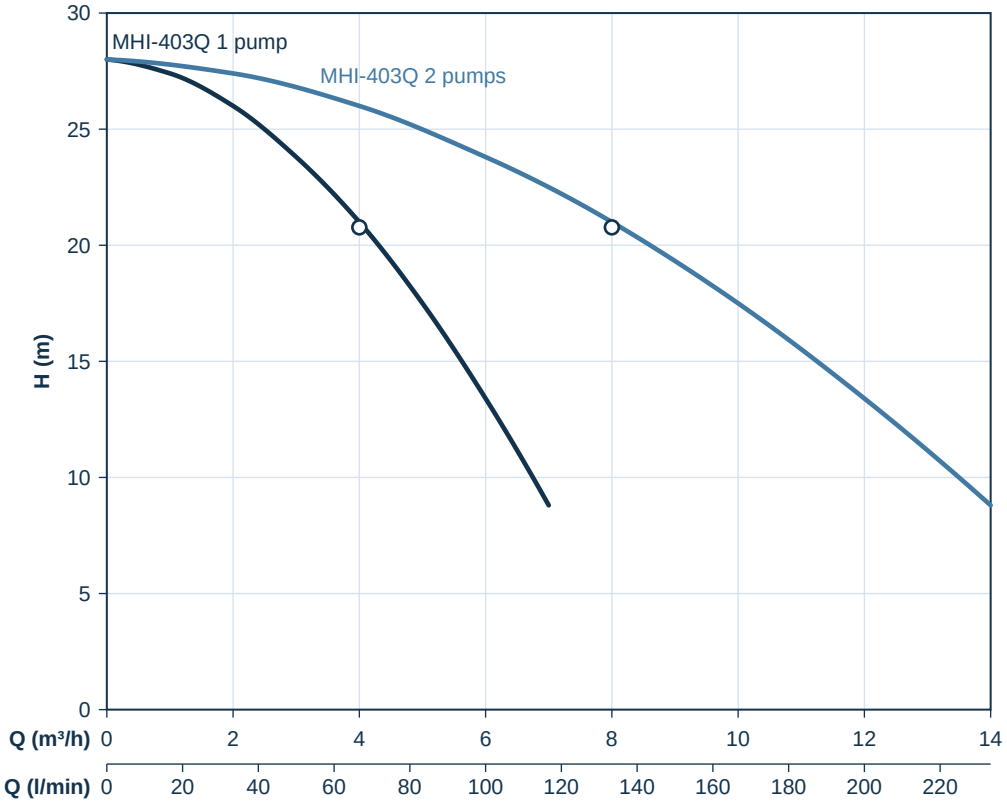
Parallel Both pumps run together for higher flow at peak demand.

TECHNICAL DATA

MODEL	PUMPS	P2 per pump kW	P2 total kW	H max m	Rated 1 pump Q m³/h / H m	Rated 2 pumps Q m³/h / H m	Q max 1 pump m³/h	Q max 2 pumps m³/h	Pump inlet / outlet DN	Pressure setting bar
MHI-403Q	2 × MHI-403	0.55	1.1	28	4 / 21	8 / 21	7	14	DN32 / DN25	2.4
MHI-1604Q	2 × MHI-1604	2.2	4.4	36	16 / 27	32 / 27	26	52	DN50 / DN40	3.0

PERFORMANCE CURVES · 1 PUMP AND 2 PUMPS IN PARALLEL

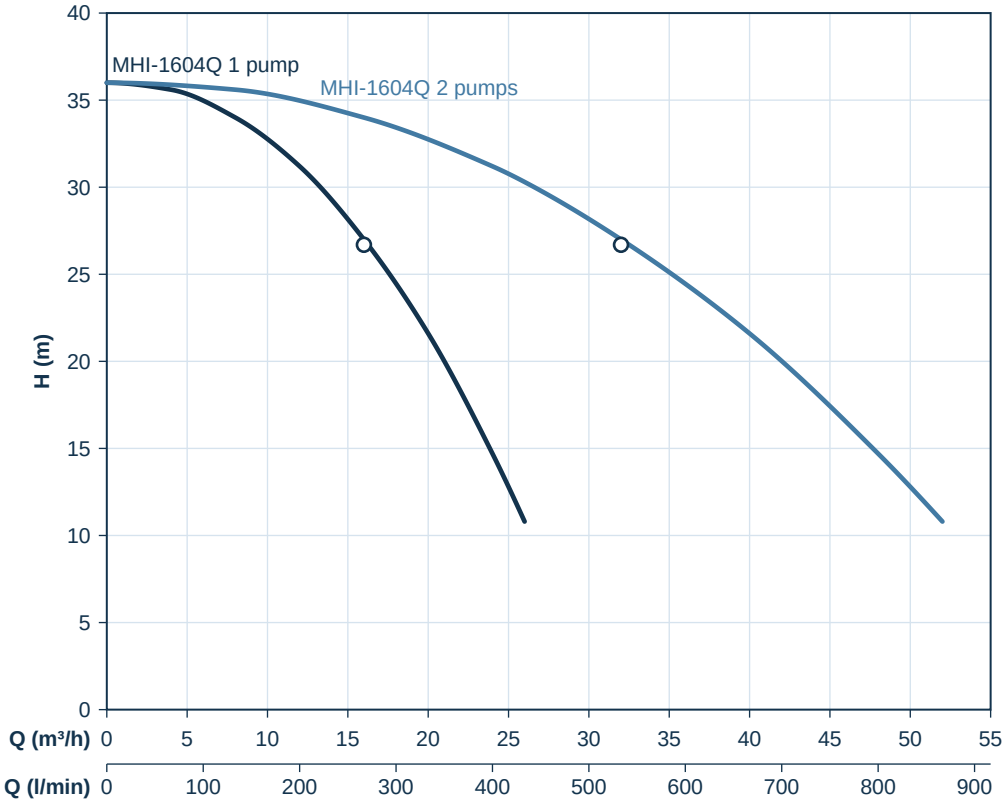
MHI-403Q 2 × MHI-403 · 2 × 0.55 kW



Q m³/h	0	2	4	6	7	8	10	12	14
Q l/min	0	33	67	100	117	133	167	200	233
1 pump	28	26	21	13.4	8.8	–	–	–	–
2 pumps	28	27.4	26	23.8	22.5	21	17.5	13.4	8.8

H (m) at Q. "–" = outside the operating range of one pump.

MHI-1604Q 2 × MHI-1604 · 2 × 2.2 kW



Q m³/h	0	8	16	24	26	32	40	48	52
Q l/min	0	133	267	400	433	533	667	800	867
1 pump	36	34	27	14.7	10.8	–	–	–	–
2 pumps	36	35.6	34	31.2	30.3	27	21.6	14.7	10.8

H (m) at Q. "–" = outside the operating range of one pump.

Curves at 50 Hz, clean water at 20 °C, kinematic viscosity 1 mm²/s. "2 pumps" = both pumps running in parallel at the same head. Circle = rated point. Typical values, not guaranteed.